

Analyzing Aviation Accident Records:

Identifying Aircraft with Minimal
Risk



Introduction:

Project Overview



This project aims to explore a complete dataset of aviation accidents from 1962 to 2023, compiled by the National Transportation Safety Board (NTSB). Through this analysis, we will identify risk factors, trends, and suggest aircraft types with minimal risk for our commercial and private aviation expansion.

Project Goals

- Are there distinct risk factors associated with increased accident rates?

- What patterns emerge over time concerning aviation incidents?.

- Which types of aircraft have experienced the most and least accidents historically?

- How can our analysis assist in making informed choices about purchasing aircraft?

Process Steps

Data Cleaning

We reduced the data to essential information for our project

Data Handling

We handled missing data through techniques of imputation

Data Analysis

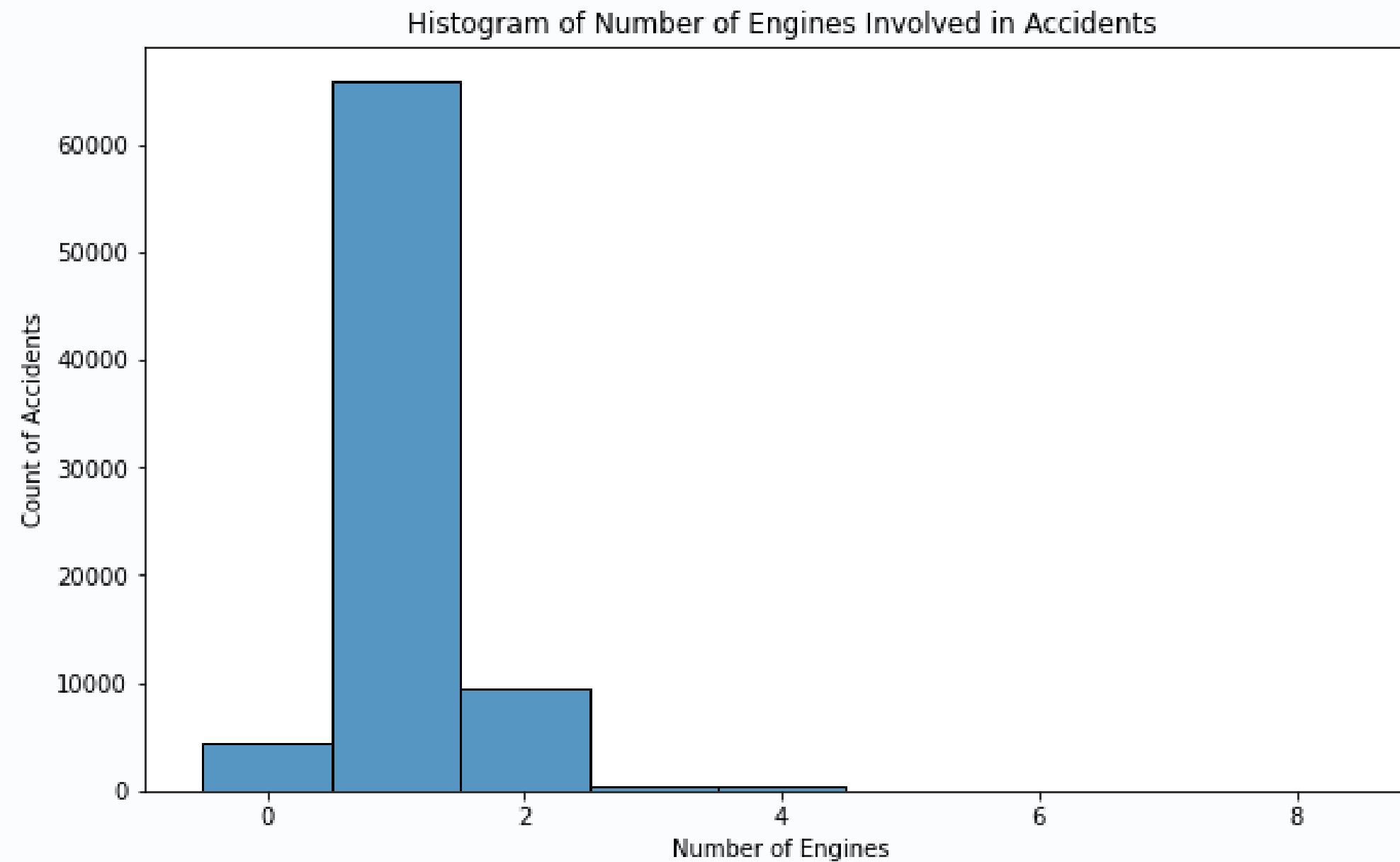
We derived valuable insights from the data

Data Visualisation

We created graphs to support our insights

Risk factors associated with increased accident rates

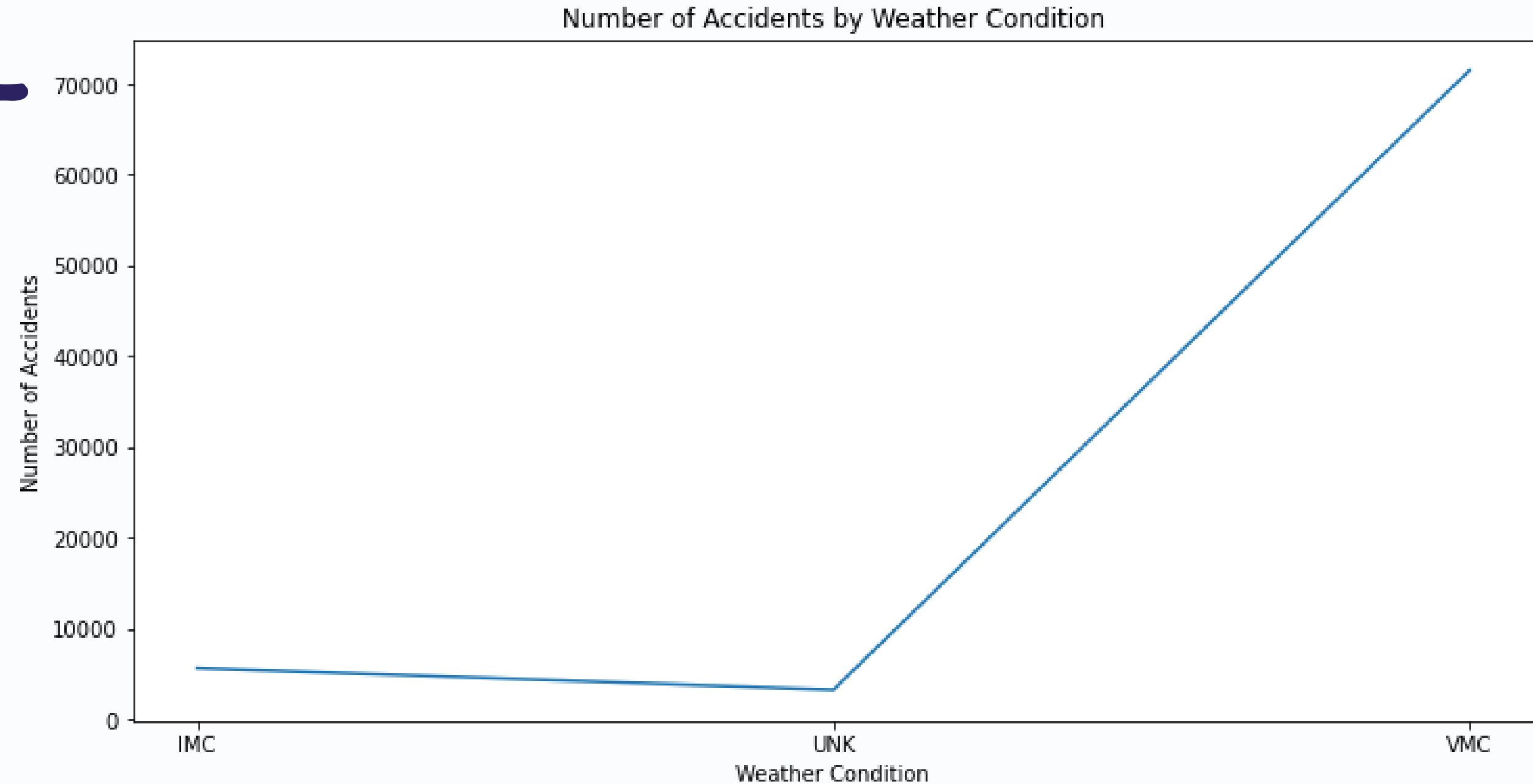
1. Number of engines in an aircraft



The risk of accidents reduces as the aircraft number of engines increases.

Risk factors associated with increased accident rates

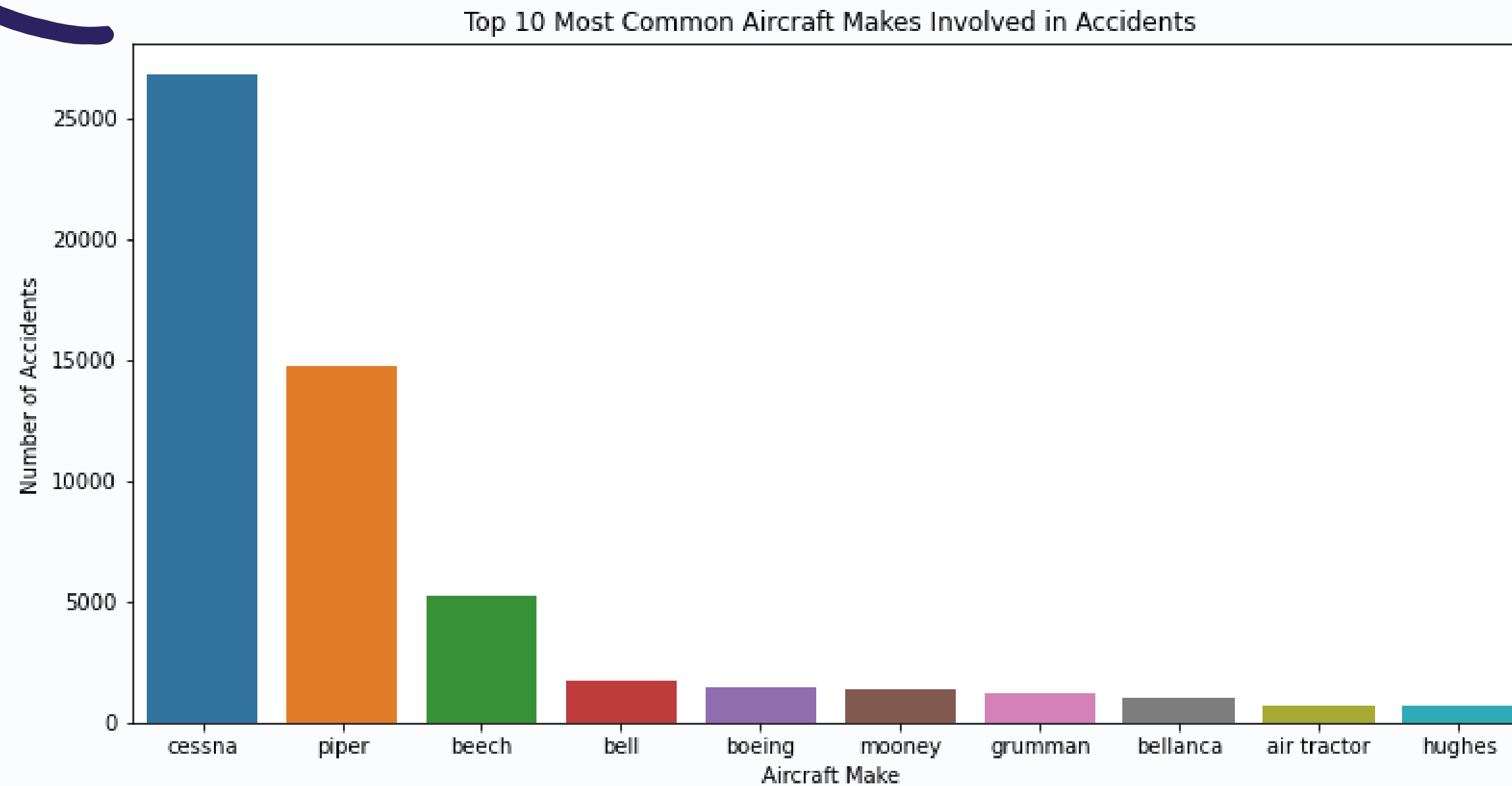
2. Weather conditions



More accidents happen during VMC conditions.

Aircrafts with the highest and lowest risk.

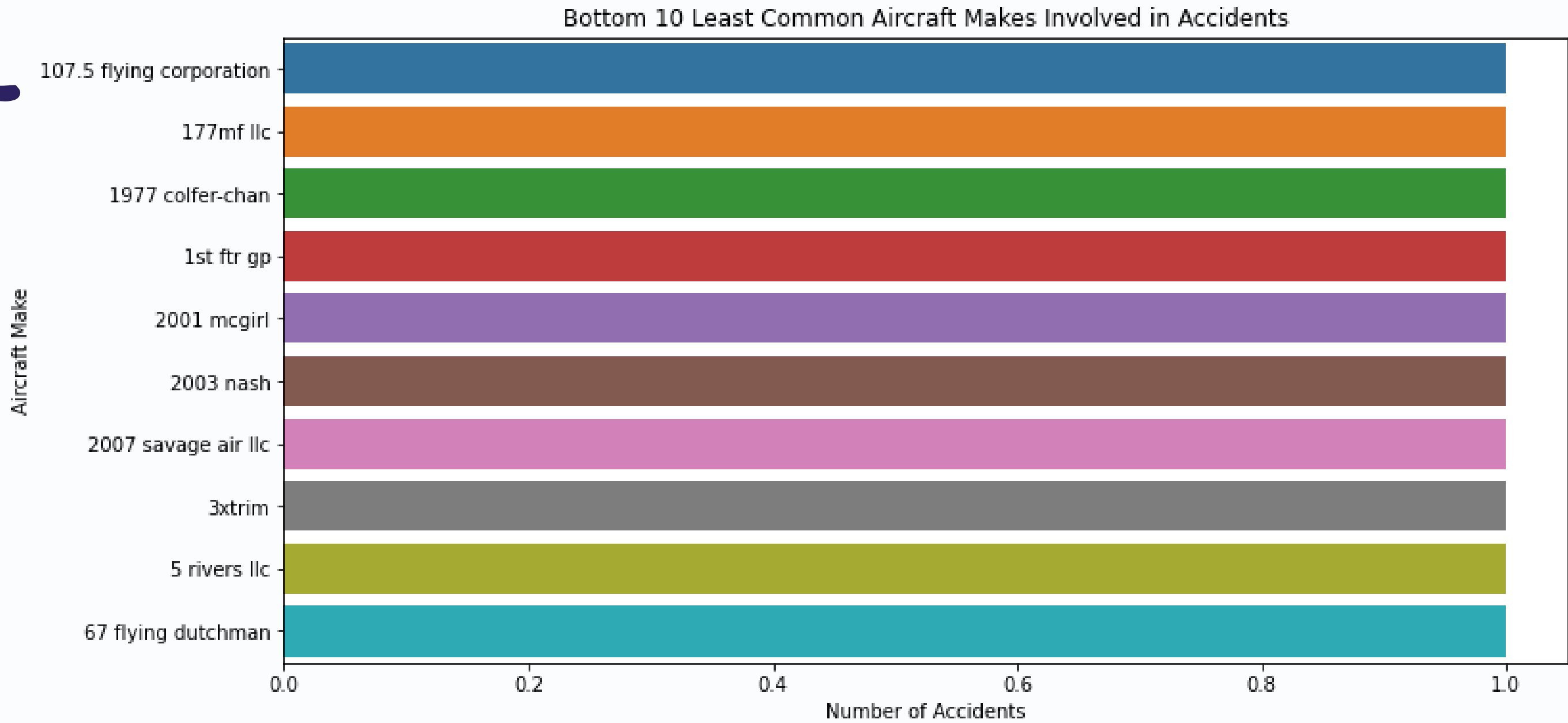
High Risk Aircrafts



There are 2 top Makes, Cessna and Piper, that have encountered the most accidents.

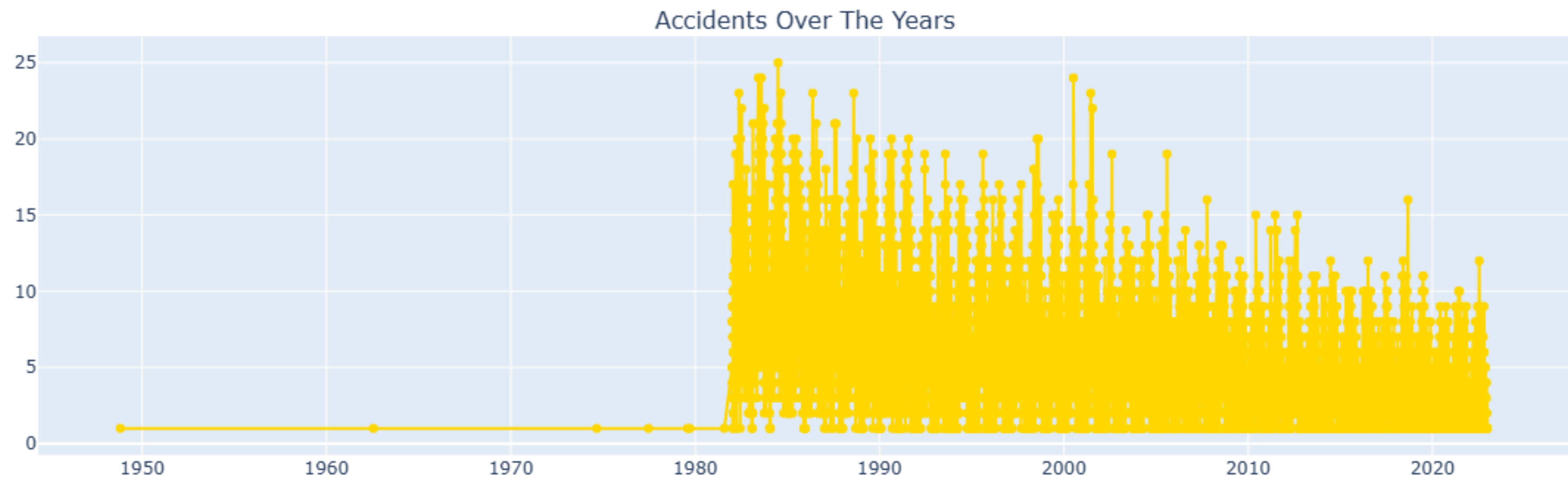
Aircrafts with the highest and lowest risk.

Low Risk Aircrafts



There are many Makes that have encountered the least accidents.

Trend Analysis of Total Aircraft Accidents Over Time



After 1982, the number of cases is decreasing every year and in 2020 there were 2.5 times fewer incidents than in 1982.

Recommendations



01

Consider aircraft types that have 2 or more engines since the accident rate is lower.

02

Prioritize the acquisition of aircrafts from makes with historically lower accident rates

03

Develop strategies to mitigate risks associated with weather conditions.

04

It is a good time to invest in aviation as the accident rates over the years has reduced

Links & Resources



Github:

<https://github.com/RGitau-coder/PROJECT-AVIATION-ACCIDENT-DATABASE-PHASE-1-/tree/main>

Tableau:

<https://public.tableau.com/authoring/AviationTableauDashboard/Dashboard1#1>



Thank You!

